

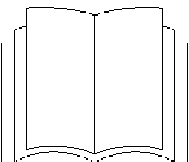


Operator's Manual for Attachment



Cutter Bar Mower

Product number: for 300-700 –series Avant loaders: A35676
for 200–series Avant loaders: A35790



Please read this manual carefully before using the equipment, and follow all instructions.

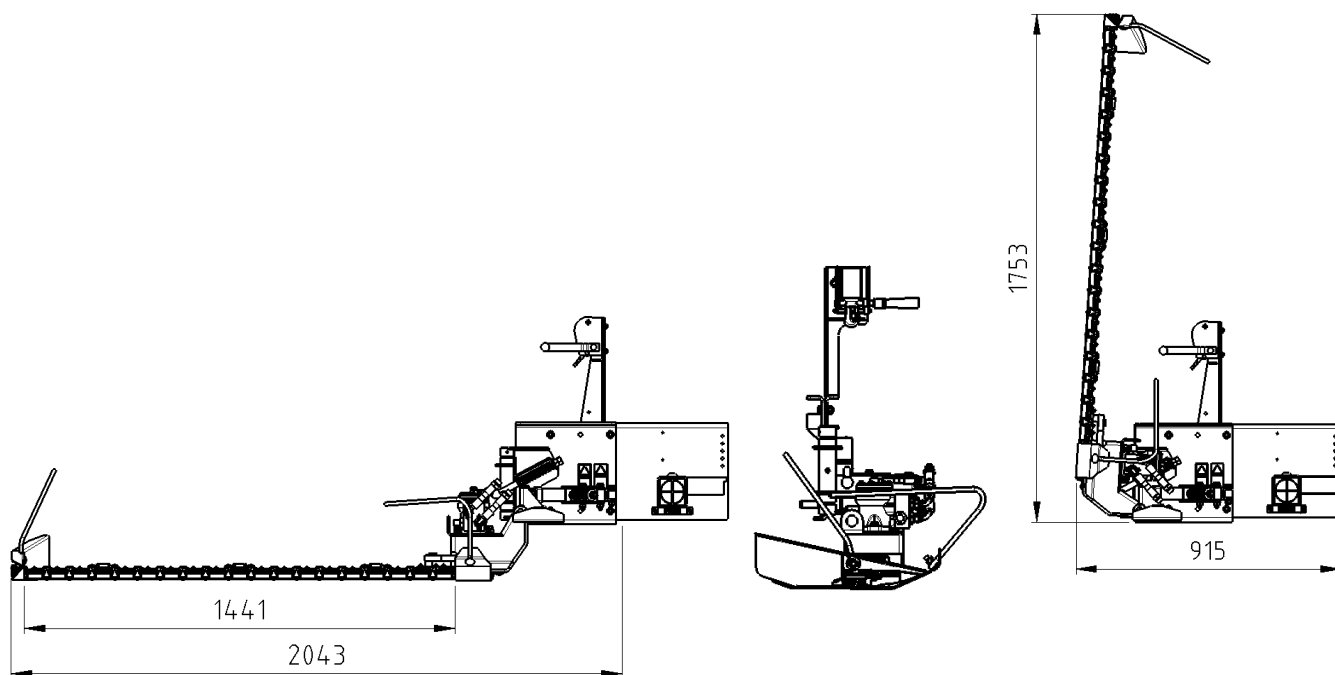
Keep this manual for later reference.

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Appendix: EC Declaration of Conformity

I. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of experience and product development. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

The purpose of this manual is to help you to:

- operate the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life

This manual uses the following warning symbols to highlight factors that must be taken into account to reduce the risk of personal injury or property damage:

	Warning: This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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NOTICE

This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader and attachment and all safety instructions before using the equipment. Other use of the equipment than described in this manual is prohibited. Keep this manual at hand throughout the service life of the equipment. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. Original instructions.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Especially the regulations governing the use of the equipment on public road areas must be observed.

2. Designed purpose of use

This hydraulically operated AVANT Cutter bar mower unit is an attachment intended to be attached on AVANT 200-700 series multi purpose loaders. The mower is a very effective attachment for cutting hay from fields, meadows and road shoulders. It makes cutting and easy collecting of even long hay possible, while allowing easy collecting of the cut hay. The attachment can also be used vertically for hedge cutting. The mower is attached on the quick coupling plate of the loader boom, giving excellent visibility to the blade bar that is on the right side of the attachment. The side reach and blade tilting make it possible to mow also from slopes. The blade guides on the blade bar support and protect the replaceable special steel blades.

Before each use, the blade is adjusted to suitable working angle with a manual winch or with the optional electric actuator system. When transporting the mower, the blade is set to its upper position. The attachment is started and operated from the driver's seat driving forward at calm speed that is adjusted according to operating and terrain conditions. The spring release mechanism of the blade bar allows the blade bar to move back when hitting an obstacle, but the operator must stay alert and observe the terrain conditions and the operation of the machine, so that hitting obstacles is avoided and the dangers of the strong and powerful cutting bar are taken into account. The attachment is intended for operators who have learned the safe operation of the loader and the attachment, and are able to notice the risks arising from using the equipment and terrain conditions. The attachment is not intended to be used by children. The mower can cut single trees of about maximum around 20 mm in diameter, but the mower is not intended for clearing areas from growing trees. After the spring release mechanism has freed the blade bar, the loader and attachment must be stopped safely, and the blade must be turned back to its operating position. Using the attachment in other ways than cutting hay or mild shrub while controlling the equipment from the driver's seat of the loader is prohibited.

The attachment has been designed to require as little maintenance as possible. The operator can perform regular maintenance tasks, but certain more demanding tasks, such as service or replacement of hydraulic components, must be left to qualified professionals. Appropriate safety equipment must be used when servicing the machine. Spare parts must be similar to original specifications, which can be ensured by using only original Avant spare parts.

Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you need spare parts or maintenance services.

Table I –Cutte bar mower - compatibility with Avant loaders

Model	216 218 220	313S 320S 320S+	419 420	520 525LPG R28 528	630 R35 635 640	745 750 760
A35676	•	-	-	-	-	-
A35790	-	(•)	•	•	•	•

When using with models marked with (•), full performance may not be achieved. Additional counterweights may also be required. For 200 series loaders, the wheel weights are recommended. To check compatibility with a model not shown in the table above, contact your Avant dealer.

3. Safety instructions for using the mower

Bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful and improper or careless use or maintenance may cause serious personal injury or property damage. Operating this attachment includes hazards that cannot be protected against with mechanical means or product design. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



Misuse or careless use, or using a mower that is in bad condition, may cause serious injuries. Familiarize yourself with the controls of the loader and coupling of the attachment at a safe area, especially how to stop the equipment in a safe manner and to make it safe. Read the following safety precautions carefully.

Read the safety instructions carefully before handling the attachment



- When attaching the attachment to the loader, **ensure that the locking pins of the loader's quick attach plate are in the lower position and that they have locked the attachment to the loader.**
- The mower is intended to be used by one adult operator at a time. Prevent others from entering the work area or getting near the loader when in use. Pay attention to the surroundings and other persons and machines moving in the vicinity. Always make sure that it is safe to move with the machine, always **keep a minimum safety distance of at least two metres from other persons.** When driving, pay attention to the contours of the terrain and, for example, holes or loose rocks, as well as other hazards, such as branches and trees reaching the driver's area.
- Never leave the driver's seat if the attachment is running, if the auxiliary hydraulic circuit of the loader has been locked on, or if the loader boom is lifted higher than one meter above the ground. Keep hands and feet away from moving parts, until the machine has been stopped following the safe stopping procedure.
- Always lower the attachment close the ground before leaving the driver's seat. Make sure that the attachment is properly supported for maintenance, cleaning, and inspections. Going under a raised attachment or boom is dangerous; the boom may lower unexpectedly due to loss of stability, equipment failure, or other persons using the controls of the loader.
- Always keep in mind that regardless to the compact size of the attachment, it is very strong, and can easily sever fingers, arms, or legs, even if the operating speed is set to low. Follow the safe stopping procedure so that any residual pressure left in the hydraulic system will not cause unexpected moving of the blades. Put the blade cover in place whenever the mower is not in use.
- Always transport the mower as low and close to the machine as possible to keep the centre of gravity low and for the best stability of the loader.
- Keep the blade bar supported against the upper spring retainer to prevent swinging and sudden movements of the bar, and to prevent hitting obstacles.
- Survey the operating area before starting the mower. Observe and mark, as necessary, any obstructions at the area, such as rocks, posts, or stay wires. Keep the limits of the machine in mind and remove obstacles or larger trees beforehand.



- Lower the blade bar using controlled motion, holding the winch handle while lowering. Operate the controls of the loader with calm and deliberate movements, avoiding unnecessary swinging of the attachment. Use the mower only when the blade bar is set to its normal perpendicular operating position, so that the spring release mechanism will hold the blade in its proper position, and allows the blade bar to swing back if necessary.
- Operate the controls of the loader only when sitting on the driver's seat. Wrong control movements create crushing and severing hazards to any person in the vicinity of the loader.
- Make sure that you use only an attachment that is in proper working condition. Perform daily inspections and read the instructions on service and maintenance. Never use the attachment if the hydraulic systems of the attachment and the loader are not completely in order.
- Do not modify the structure of the attachment in a manner that would affect its safety. Modifying the blade system or their fixing, or hindering the spring release mechanism, may prevent safe use of the attachment and cause a hazard to the operator.
- Ensure that the loader is equipped with necessary safety equipment, and that they are in working condition. Seat belt must be used. If there are specific hazards at the operating area, such as dust or hazard of falling objects, take necessary steps to protect from those.
- Also read the safety instructions shown in the operator's manual of the loader.



Never perform any cleaning, maintenance, service, or adjustment tasks if the loader engine is running or if the auxiliary hydraulics is locked on. Blade bar that has become stuck may move unexpectedly when removing a blockage because of residual pressure left in the hydraulic systems may cause cutting motion, creating severing hazard to any person near the blade bar. Follow the safe stopping procedure shown below.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised boom or mower. Always lower the attachment near the ground before leaving the driver's seat.

Safe stopping of the equipment, before going near the attachment:



Always stop the mower following the safe stopping procedure before going near it. Safe stopping procedure prevents unintentional movements of the attachment. Safe stopping procedure:

- Stop the attachment and the loader, disengage the auxiliary hydraulics lever of the loader.
- Lower the boom and tilt the attachment so that it is on the ground or just above ground.
- Shut down the loader engine and lock the parking brake.
- Release residual pressure from the hydraulic system.
- Prevent starting of the machine, remove ignition key.

Remember to wear proper protective clothing:


- The noise level at the driver's seat may exceed 85 dB(A). Wear hearing protection while working with the loader.



- Wear safety boots while working with the loader.



- Wear protective gloves and safety goggles when handling hydraulic components. Wear protective gloves when handling the blades of the attachment.

4. Technical specifications

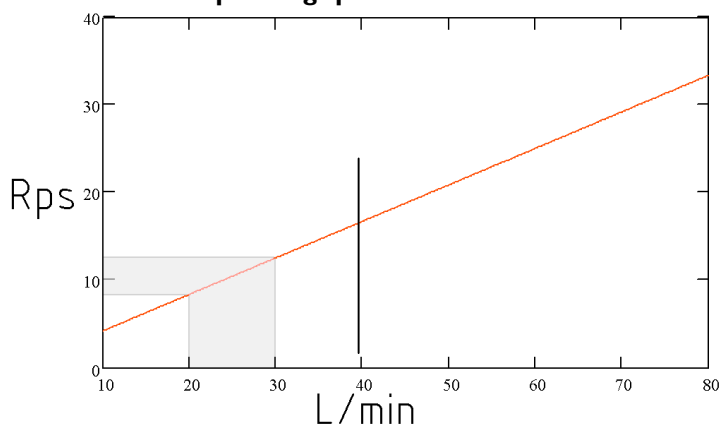
Table 2 – Avant Cutter bar mower – technical specifications

Product number:	A35676 A35790
Working width:	1400 mm
Side reach: (from middle of quick attach plate)	2040 mm
Reach drawing:	See Chapter 6.2
Dimensions:	See Page 2
Weight:	70 kg
Blade bar:	19 pcs 65810, teethed
Nominal hydraulic flow:	20-30 l/min
Maximum recommended hydraulic flow:	40 l/min
Maximum operating pressure:	22,5 MPa (225 bar)
Suitable AVANT loader:	A35676 300-700 series A35790 200-series

Available options:

- Electric blade bar control A35975
- Long hay mowing kit A416962

Ask for the availability of the options from your dealer. For installation services, refer to service point as needed.

Graph I – Rotating speed of Cutter bar mower according to hydraulic oil flow and recommended operating speed


4.1. Safety labels and main components

Listed below are the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information and they help to identify and remember the hazards related to the equipment. Replace any damaged or missing warning label with a new one.



A46771



A46772



A46803



A46749

AVANT Avant Tecno Oy Ylötie 1 FIN-33470 YLÖJÄRVI	
TYPE	No.
kg l/min	
Max l/min,	MPa (bar)
2014 S	Made in EU www.avanttecno.com
CE	

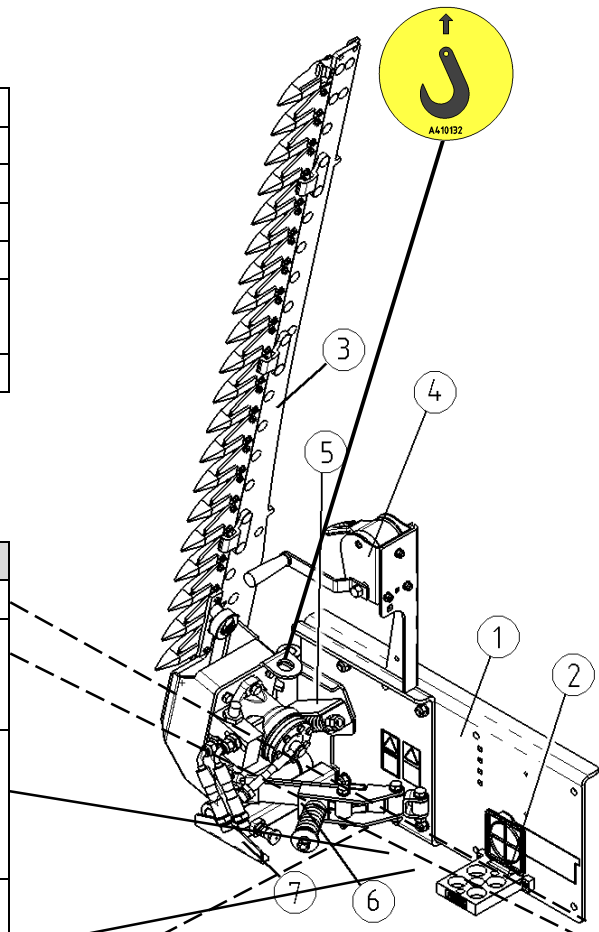
Attachment ID plate

Table 3 – Main components of Cutter bar mower

1	Frame with Avant quick coupling plate
2	Multiconnector parking stand
3	Blade bar, 19 blades
4	Blade bar lowering winch
5	Blade bar upper position spring retainer
6	Spring release mechanism and compression spring
7	Storage support

Table 4 – Decal warning messages and locations

Decal	Message
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or the loader boom, keep a safe distance from lifted equipment.
A46803	Crushing hazard, moving blade bar and joint. Lower the blade bar with controlled motion and keep hands and feet away from hazard areas.
A46709	Severing hazard, sharp and strong blades. Follow safe stopping procedure.
	Identification plate of the attachment



5. Assembly of the Cutter bar mower

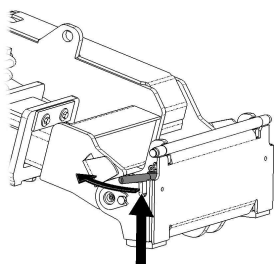
The attachment is ready for use as delivered. Attach the attachment and its hydraulic hoses to the loader as instructed below.

5.1. Coupling the attachment to the loader

Coupling the attachment to the loader is quick and easy, but it must be done carefully. The attachment is fixed to the loader boom by using the quick attach plate and the counterpart on the attachment. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide towards the driver along a raised boom. For this reason, the attachment must not be raised over one metre under any circumstance if it has not been locked. To prevent hazardous situations, always follow the attachment coupling instructions provided in the following pages. Also remember the safety instructions shown in Chapter 3 and additional information shown in the operator's manual of the loader. The attachment is mounted to the loader as follows:

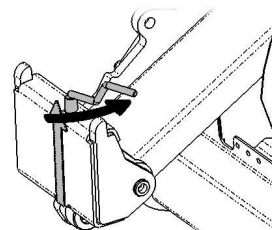
300-700 series loaders:

Step 1:

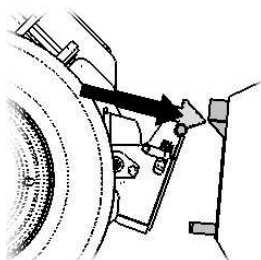


- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.

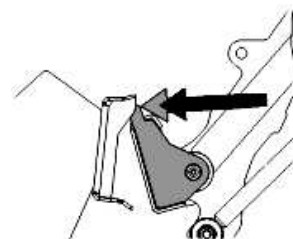
200 series loaders:



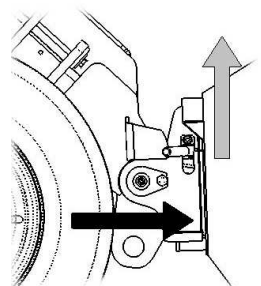
Step 2:



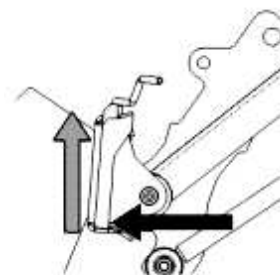
- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this. Align the upper pins of the loader's quick coupling plate so that they are under the corresponding brackets of the attachment.
- Depending on the storage position of the mower and the reach capability of the loader, it may be necessary to assist the mounting of the attachment manually. See Chapter 5.4 for correct uncoupling procedure and storage position.



Step 3:

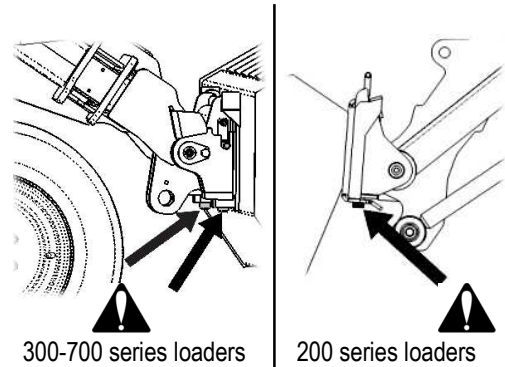


- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually or lock the hydraulic locking. Always check the locking of pin / pins.





An attachment that has not been completely locked to the loader may fall or tip over on the boom or operator, or under the loader during driving, causing loss of control of the loader. Before moving or lifting the attachment, make sure that the locking pins are in the lower position and come through the fasteners on the attachment on both sides. Never move or lift an attachment that has not been locked.



300-700 series loaders

200 series loaders



Excessive tilting or lifting of an unlocked attachment increases the risk of tipping the attachment over. Do not use the automatic locking of the locking pins when the attachment is lifted more than one metre from the ground. If the locking pins do not return to the normal position when tilting, do not tilt or raise the attachment any more. Lower the attachment on the ground and secure the locking manually.

5.2. Connecting and disconnecting the hydraulic fittings

On 300-700 series loaders hydraulic hoses are connected using the multi connector system. If you want to change the type of the hydraulic couplings, contact your Avant dealer or service point for instructions or installation services. On 200-series loaders only conventional quick couplings are available.



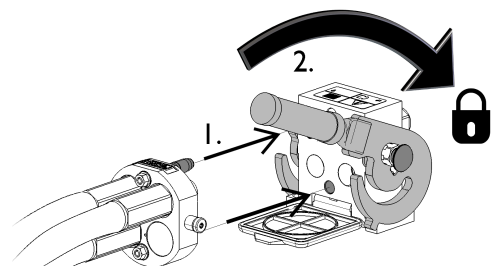
Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious injuries or burns.

NOTICE

Keep all fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting the multi connector system:

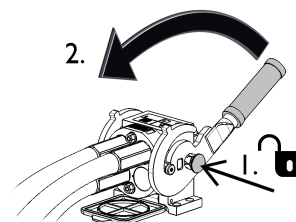
1. Align the pins of the attachment connector with corresponding holes of the loader connector. The multi connector will not connect if the attachment connector is upside down.
2. Connect and lock the multiconnector by turning the lever towards the loader.



The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.

To disconnect the multi connector system, turn off the auxiliary hydraulics of the loader, and:

1. Push the knob on the side of the connector to unlock.
2. Turn the lever to disconnect.



Conventional quick couplings (alternative couplings for some loader models):

Before connecting or disconnecting the standard quick couplings, the residual pressure must be released as shown below. The conventional quick couplings will not connect, if there is pressure in the hydraulic system.

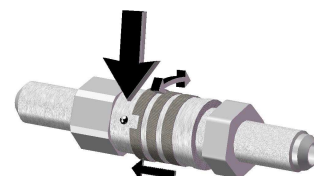
To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the fitting equipped with a coloured cap is connected to the corresponding fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



NOTICE

Make sure that all three couplings are properly connected, otherwise the hydraulic motor of the mower may get damaged.

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. There is a small groove in the collar of all female fittings, and a small ball near the collar. When connecting or disconnecting, the groove and ball must be aligned. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The collar may also turn accidentally to the locking position, making it impossible to connect or disconnect the couplings.



Releasing residual hydraulic pressure:

To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader to it's extreme positions before disconnecting the couplings. If the fittings will not connect to the loader, the residual pressure must be released. The residual pressure must be released before handling the blades of the attachment and before storage of the attachment.

Disconnecting the hydraulics:

Before disconnecting the fittings, place the attachment on solid and level surface, as described in chapter 5.4. Follow safe stopping procedure.

NOTICE

Always disconnect the hydraulic fittings first before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent dirt from entering the hydraulic system.

5.3. Electrical connection, electric blade bar control (optional equipment)

The mower can be equipped with an electric actuator, that replaces the manual winch for lowering the blade bar. In this case, the electric harness of the attachment must be connected to the loader. There are two options for the electrical connection:

- Electric blade bar lowering can be controlled with the attachment control switch pack, that is available as an option to 400-700 series loaders. When using the switch pack, the electric cable of the loader is connected to a plug near the hydraulic fittings.
- Alternatively, a cable harness A35358 equipped with a 2-way switch, that is delivered with the mower, can be used. Instructions concerning the loader plug are provided in the loader manual. Select the location of the switch in the cab to ensure that it is easily accessible but will not be used inadvertently.

Route the selected cable in a way that it cannot get stuck on anything or stretch when the equipment moves. With the help of the attachment control switch pack available for the loader, the attachment control switches are easily accessible at the end of the boom control lever. See instructions about using the electric actuator in chapter 6.2.

5.4. Uncoupling the attachment and storage

Storage place that is protected from rain and direct sun is the best for an attachment. Uncouple the attachment only on level and firm surface, which can support the weight of the attachment steadily. To avoid storing the attachment directly on the ground for long periods of time to avoid damage from corrosion, place the attachment on a pallet, for example.

NOTICE

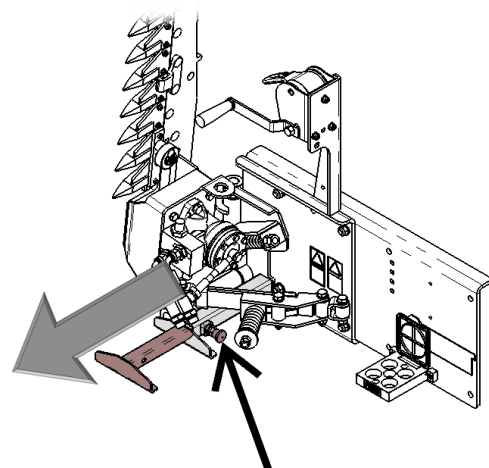
Avoid storing the attachment directly on the ground for long periods of time to avoid damage from corrosion. Clean and service the attachment before long storage periods as shown in chapter 7.

Correct storage position:

At the front of the mower, there is a support stand, which must be extended to prevent the mower from tipping forward.

Uncoupling the mower:

1. Select a level and firm storage area that can support the mower standing upright.
2. Stop the attachment following the safe stopping procedure. Uncouple hydraulic fittings and electric harness.
3. Fit blade cover.
4. Turn the blade bar up against the retainer spring.
5. Extend and lock the support stand shown in adjacent figure.
6. Lower the attachment completely down in upright position and unlock locking pins. Tilt the quick coupling plate carefully and manoeuvre the loader away.



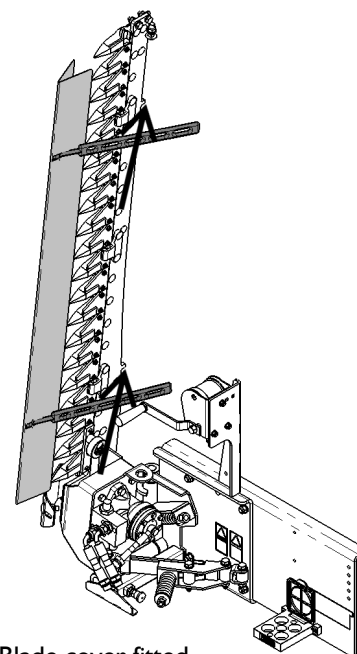
A loader that is not equipped with a telescopic boom:

If the loader is not equipped with a telescopic boom, and the mower is stored directly against the ground, the loader may not reach the brackets of the mower's quick coupling plate. In this case, a separate lift can be used, or the mower can be tilted slightly manually holding from its blade bar, so that the quick coupling brackets will set above the loader's coupling plate. Before handling the mower manually, drive the loader as close to the quick attach plate of the mower as possible, and tilt the attachment plate of the loader forward to minimise the needed manual tilting.

Blade cover:

The blade cover protects both the attachment and any person near the attachment. Always fit the blade cover after use and do not remove until work can be started.

Clean the blade before storage. Allow the blade to dry before fitting the cover, if storing for longer periods of time.



Blade cover fitted

6. Instructions for use

Check the attachment and the operating area once more before starting to work. Quick inspection of the mower before starting to use it is part of ensuring safety and the best performance of the equipment. Practise the use of the mower and its controls in an open and safe area. If you are unfamiliar with the loader to be used, it is recommended to practice its use without any attachments.



Beware of the sharp and strong blade bar. Start the mower only after it has been set to working position and work can be started. Always stop the attachment before leaving the driver's seat and keep a safe distance to others. Follow safe stopping procedure always when it is necessary to go near the blades or when handling the blade bar, i.e. cleaning, height adjustment, or uncoupling the attachment. Always fit the blade cover when the attachment is not in use.



6.1. Checks before use

- Check the general condition of the attachment and the loader, and check for possible hydraulic oil leaks. The attachment must not be used, if there is a fault in the hydraulic system of the loader or the attachment.
- Ensure that the loader and the attachment are used in a safe and intended manner. Before starting mowing, check the operating area and remove or mark objects affecting the safety of mowing or the stability of the loader. Remember correct working methods and avoid leaving the driver's seat unnecessarily.
- Check the condition of the blade lowering wire, so that the blade bar will remain in its intended position. Do not use the machine if the blade bar is lowered against the ground.
- Check that the blade bar is intact and clean. Remove any trash stuck on the blade bar. Check visually that the blades are intact and can move freely.
- Check that the blade bar is in its normal perpendicular operating position supported by its spring release mechanism, and has not tilted backwards. See instructions about the blade spring release mechanism from chapter 6.3.
- Operate the attachment or any controls only when sitting on the driver's seat.

NOTICE

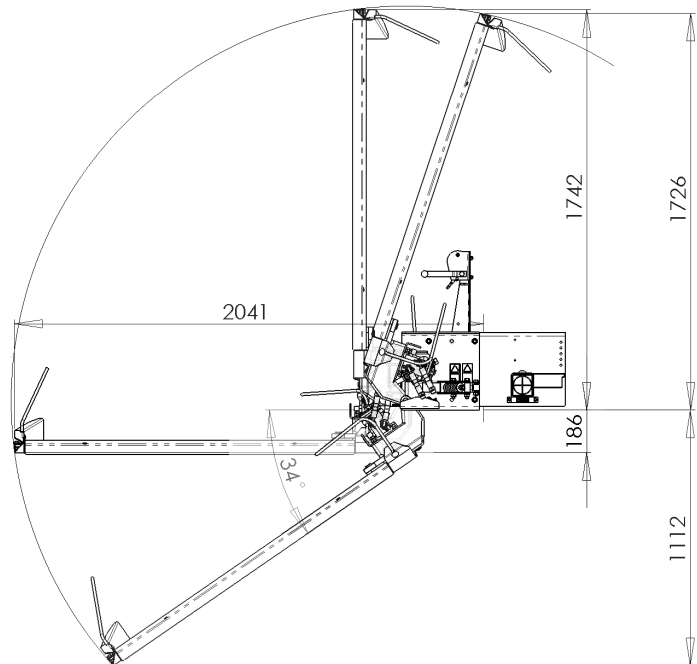
The loader should be equipped with a hydraulic oil cooler when using the mower. Overheating of the oil decreases performance and affects the service life of the hydraulic components of both the attachment and the loader. Regularly clean the loader oil cooler core; refer to the loader manual for more instructions.

6.2. Operating the mower

Working position and reach of blade bar:

The blade can be set according to operating area as shown in adjacent figure. In near vertical positions a spring retainer will hold the blade in desired position, preventing swinging of the blade.

The blade bar should be set to near the surface of the ground, but not completely leaning on the ground. The lifting cable must be connected and tensioned when mower is in use.

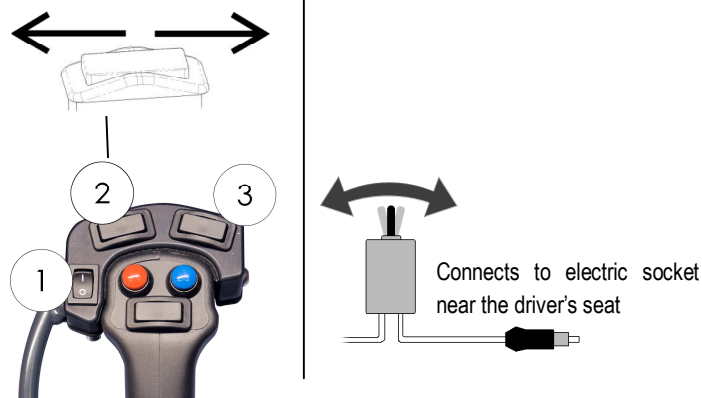


Reach diagram of the blade bar

Turning the blade bar to operating position:

- Before handling the manual winch, stop the attachment following safe stopping procedure. Lower the blade bar with controlled motion, holding the winch handle throughout lowering.
- When using the electric blade bar lowering, the actuator is controlled with switch number 2 of the attachment control switch pack, or by using the 2-way switch delivered with the attachment.

Electric lowering of blade bar



Starting the mower:

The mower is started by turning the auxiliary hydraulics control lever of the loader to its locking position. Correct running speed of the loader engine depends on operating conditions and driving speed. The mower can cut also when it runs slowly, but for the best results, the loader should generate hydraulic oil flow of about 20-30 l/min. On most loaders this is achieved when the loader engine is run at about half of the maximum rpm, using one hydraulic pump. If needed, check the hydraulic oil output of the loader from the operator's manual of the loader.

During operation:

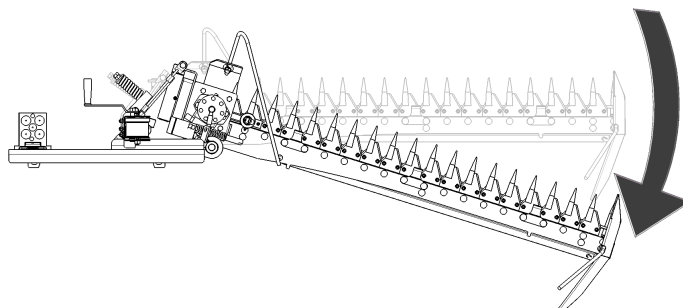
Operating on inclined terrain and slopes requires extra caution. Watch for holes, bumps, rocks and other possible obstacles, that may be poorly visible. Uneven terrain may tip over the loader. Use low speed on slopes and do not make sudden changes in speed or direction. Try to drive straight up or down a hill, not across, to avoid excessive side tilting. If the loader stops while driving up a steep slope, first stop the attachment and then reverse a little. Keep the articulation of the loader frame in as straight position as possible. Be careful when operating near ditches or embankments as the machine could slide or turn over if an edge caves in. Do not lift the attachment high on inclined terrain.

6.3. Blade bar spring release system

To protect the attachment and the operator the blade bar will swing back if the blade bar hits an obstacle. The system will not protect the mower or it's blades against overload, and the attachment may get damaged regardless of the release system.

The spring release system should work so that it will allow the blade bar to tilt back when moderate force is applied to the tip of the blade bar.

When the spring release system allows the blade bar to tilt back, stop the loader immediately and stop also the attachment. The blade bar can be turned back to operating position by operating the loader boom, or by turning it back manually. Remember to follow safe stopping procedure before going near the blades.



Movement of the blade bar, spring release system



The spring release system does not replace careful and cautious driving style, and the system can't fully remove hazards related to sudden stopping of the machine. Always use seat belt and drive slowly, and remove or mark any obstacle in the operating area so that collisions can be avoided.

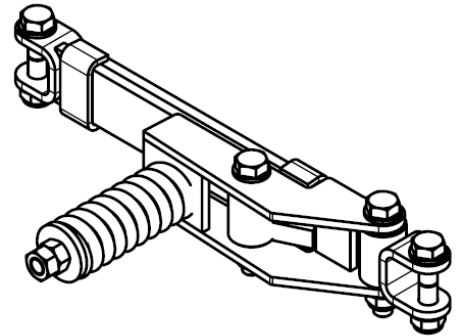


Operate the mower only when the blade bar is in its normal operating position retained by the spring release mechanism, perpendicular to the driving direction. Check the operation of the spring release system regularly. Blocking the release system completely will expose the operator to hazards related to sudden stopping of the loader, make sure that the system is in proper working condition.

If the system does not work as desired, allowing the blade bar to turn too easily, it can be adjusted by tensioning its spring. The system should be kept clean. Some oil can be sprayed in between the sliding surfaces. Normally, the system shouldn't give in when pulling the blade bar manually



The spring of the release system is under compression tension. Careless handling of the system or its spring can cause the spring to be thrown. Leave the outer spring retaining nut in its place.

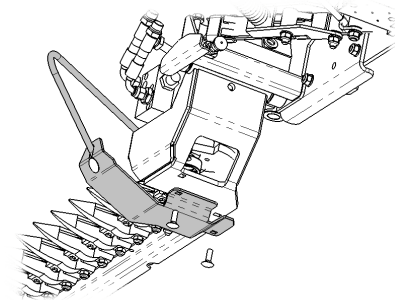


Spring release mechanism of the blade bar

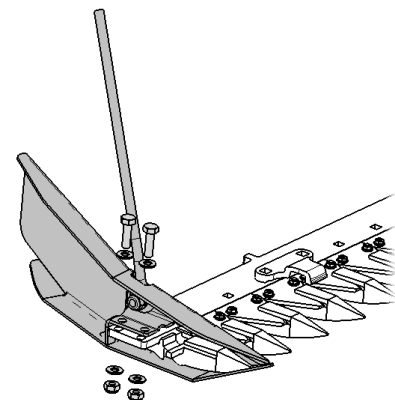
6.4. Long hay mowing kit (optional equipment)

When cutting long hay, the mower can be equipped with long hay mowing kit, which reduces entangling of hay onto the blade bar, and will guide the clippings for more even performance. The kit also protects the end of the blade bar, making it recommendable to install the kit whenever the blade may hit the ground often.

Long hay mowing kit includes guide/skid sets that are installed on both ends of the blade bar.



Long hay mowing kit, base of the blade bar



Long hay mowing kit, tip of the blade bar

7. Maintenance and servicing

The attachment has been designed to be as maintenance free as possible. Continuous maintenance of the mower includes careful cleaning and regular lubrication. Also, the condition of the hydraulic hoses and components must be monitored regularly. Because of the crushing hazard caused by the loader boom and the severing hazard of powerful cutting blades, all maintenance procedures must be made when the attachment is lowered completely against the ground and when the residual pressure of the hydraulic system has been released. For the best stability of the mower, it should be kept locked on the loader during maintenance.

Spare parts are available from your nearest AVANT representative. Please contact your nearest Avant dealer if you have any additional questions, or if you need any maintenance services.



Ensure that the attachment is properly supported also when it is connected to the loader. All maintenance procedures must be performed while the attachment is lowered on the ground, the loader engine is shut down and the residual pressure is released. The loader boom may lower by itself causing crushing hazard.



Before handling the blades, the attachment must be stopped following the safe stopping procedure, so that the blades can't unexpectedly make cutting movement. Also, wear protective gloves when handling sharp blades.

7.1. Inspecting hydraulic components

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Do not use the equipment, if you have discovered a leak in the hydraulic system of the attachment or the loader. Leaking hydraulic fluid may penetrate skin and cause serious injuries. Seek medical attention immediately in case hydraulic fluid penetrates the skin. Wash any part of body that has been in contact with hydraulic oil carefully with water and soap. Hydraulic fluid is also harmful to the environment and any leak to the environment must be prevented. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Operate the attachment only with type of hydraulic oil that is accepted for use in Avant loaders.



Never handle hydraulic components when the hydraulic system is pressurized, since a fitting may break or become loose and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



Check hoses visually for cracks or abrasions. If there are signs of leaks, to check a component, hold up a piece of cardboard in the area where a leak is suspected. Do not use hands to search for leaks. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. Check the routing of the hoses; adjust the hose clamps to avoid abrasion to the hoses. The hoses have limited service life, which is marked on the hoses. After the date marked on the hoses, they should be replaced with new ones.

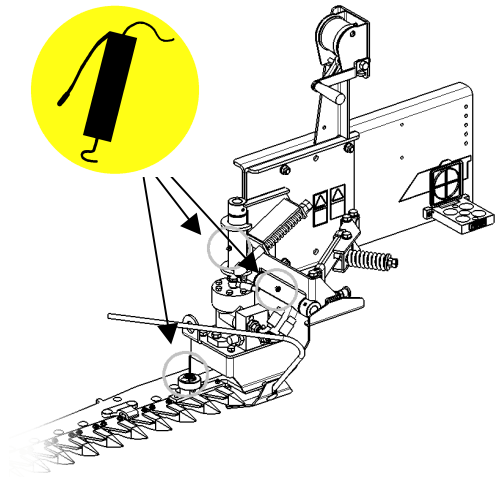
Finding any fault means that the hydraulic hose or component must be replaced and the equipment must not be used until it is repaired. Spare parts are available from your nearest AVANT retailer or authorized service point. Leave the repair work to professional service technicians, if you don't have adequate knowledge and experience about hydraulic assemblies and how to perform the repairing safely.

7.2. Cleaning and lubrication

Clean the attachment regularly. Pressure washer and mild detergent can be used. Spraying on hydraulic or electric components or the labels on the attachment should be avoided. Also avoid spraying directly on the winch cable. Allow the machine to dry and lubricate before fitting the blade cover.

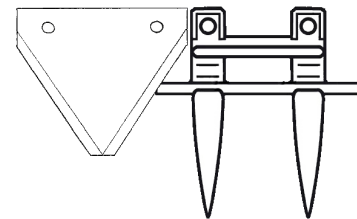
Before long storage periods touch up paint where needed, add grease the lubricating points and apply thin oil on the blade bar to prevent rust damage. There are a total of three lubricating points, located in vertical joint of the blade bar, hinge of the spring release system, and at the bearing of the blade moving rod, on top of the blade bar. Lubricant should be added to the bearing after every 8 hours of use, and less frequently to the hinges. The hydraulic motor end of the rod is maintenance free.

The winch gears may be oiled lightly as needed, but applying oil or grease on the cable should be avoided, as it would cause the cable to collect dirt shortening its service life. The electric actuator system is maintenance free.



7.3. Mower blades

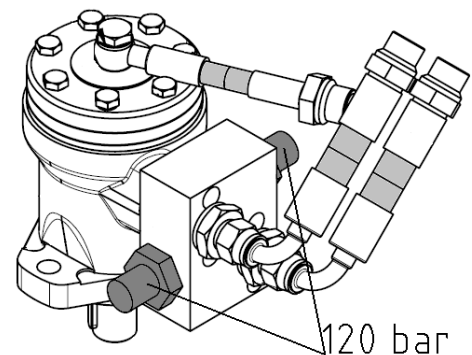
There are a total of 19 blades number 65810, whose teathed edges help to retain good cutting properties over a long time. Damaged blades can be replaced, and also blade guides should be checked and replaced as necessary. The blades must slide smoothly along the blade guides, and if a blade or guide gets damaged, they must be replaced before continuing use.



Blade 65810 and blade guide

7.4. Hydraulic motor settings

To protect the blades, the hydraulic motor is assembled with a pressure relief valve, set at 120 bar. If needed, the pressure can be checked and adjusted. Original value must not be exceeded.



Pressure relief adjustment of hydraulic motor

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EU samsvarserklæring
EF-overensstemmelseserklæring
EC Declaration of Conformity
EG-Konformitätserklärung
Déclaration de conformité CE



Valmistaja / Tillverkare / Produsent / Producent / Manufacturer / Hersteller / Fabricant:
Osoite / Adress / Adresse / Address / Adresse:

AVANT TECNO OY
Ylötie 1
33470 YLÖJÄRVI
FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu /

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats /

Vi erklærer herved at produktet som er oppgitt under er i samsvar med forskriftene i Maskindirektivet (direktiv 2006/42/EC med endringer). Følgende harmoniserte standarder har blitt anvendt /

Vi erklærer herved, at nedenstående produkter er i overensstemmelse med bestemmelserne i maskindirektivet (direktiv 2006/42/EF indeholdende ændringer). Følgende harmoniserede standarder er anvendt /

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied /

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt /

Nous déclarons par la présente que les produits mentionnés ci-dessous sont conformes aux exigences en matière de sécurité et de santé de la directive relative aux machines (directive 2006/42/CE, avec ses modifications) Les normes harmonisées suivantes ont été appliquées :

SFS-EN ISO 12100, SFS-EN ISO 4413

Kuvaus / Beskrivning / Beskrivelse / Beskrivelse / Description / Beschreibung / Description :

Hydraulitoiminen palkkiniittokone; Avant-kuormaan työlaite

Hydraulisk slåtterbalk; arbetsredskap för Avant lastare

Hydraulisk fingerslåmaskin; redskap for Avant minilastere

Hydraulisk fingerklipper redskab til Avant Minilæssere

Hydraulic Cutter bar mower; attachment for Avant loaders

Hydraulische Balkenmähwerk; Anbaugerät für Avant Radlader

Barre de coupe a doigts à commande hydraulique destinée à être utilisée avec les chargeurs Avant

Mallit / Modeller / Modeller / Modeller / Models / Modellen / Modèles:

Avant	216, 218, 220	313S, 320S, 320S+, 419, 420, 520, 525LPG, R28, 528, 630, 635, 635, 640, 745, 750, 760
Palkkiniittokone Slåtterbalk Fingerslåmaskin Fingerklipper Cutter bar mower Balkenmähwerk Barre de coupe a doigts	A35790	A35676

1.9.2014 Ylöjärvi, Finland



Risto Käkelä,

Toimitusjohtaja / Verkställande direktör / Administrerende direktør /
Managing Director / Geschäftsführer / Directeur général



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